

1 Notations

- $\mathbf{x}$: vector
- $\mathbf{X}$: matrix

Suppose that the $\mathbf{X}$ consists of vector components $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. Then,

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix}$$